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3.1.2 Price Determination in a Competitive Market

Demand

Demand is the quantity of a good or service that consumers are *willing and able* to buy at a given price in a given time period. Both conditions must be met — wanting a Ferrari is not demand unless you can also afford one.

Individual Demand vs Market Demand

Individual demand is the demand of a single consumer. **Market demand** is the horizontal summation of all individual demand curves in the market — it shows the total quantity demanded by all consumers at each price. In practice, when we draw demand curves, we almost always mean market demand.

The Law of Demand

There is an **inverse (negative) relationship** between price and quantity demanded, ceteris paribus. As price rises, quantity demanded falls; as price falls, quantity demanded rises. This gives the demand curve its characteristic downward slope from left to right.

Three reasons why the demand curve slopes downward:

- **Income effect:** When the price of a good rises, consumers' real income (purchasing power) falls — they can afford less with the same money. So they buy less of the good. For example, if petrol prices double, a driver's weekly budget stretches to fewer litres.
- **Substitution effect:** When the price of a good rises, it becomes relatively more expensive compared to alternatives. Consumers switch to cheaper substitutes. For example, if the price of Coca-Cola rises, some consumers switch to Pepsi or own-brand cola.

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- **Diminishing marginal utility:** The additional satisfaction (utility) from each extra unit consumed decreases. Consumers will only buy extra units if the price falls to reflect this lower marginal benefit. The first slice of pizza gives great satisfaction; the fifth gives much less — you'd only eat it if it were much cheaper.

The Demand Curve

Figure 1: The demand curve — as price falls from P_1 to P_2, quantity demanded extends from Q_1 to Q_2

Movements Along vs Shifts of the Demand Curve

This is one of the most important distinctions in economics — getting it wrong loses marks in nearly every exam:

	Movement Along the Curve	Shift of the Curve
Caused by	A change in the good's **own price**	A change in any **non-price factor**
What happens	We move from one point to another *on the same curve*	The entire curve moves left or right

Terms used	**Extension** (price falls, Qd rises) or **contraction** (price rises, Qd falls)	**Increase in demand** (shift right) or **decrease in demand** (shift left)

Exam Tip: Never say "demand increases" when you mean "quantity demanded increases." An increase in *demand* means the whole curve shifts right. An increase in *quantity demanded* means a movement along the existing curve due to a price fall. This distinction is tested heavily.

Non-Price Determinants of Demand (Demand Shifters)

These factors cause the entire demand curve to shift:

Factor	Effect on Demand	Example
Income	Normal goods: income ↑ → demand ↑ Inferior goods: income ↑ → demand ↓	Rising incomes increase demand for organic food (normal good) but decrease demand for supermarket value ranges (inferior good)
Price of substitutes	Substitute price ↑ → demand ↑	If Uber raises prices, demand for black cabs increases
Price of complements	Complement price ↑ → demand ↓	If printer prices rise, demand for ink cartridges falls
Tastes & preferences	Favourable change → demand ↑	A viral TikTok trend increases demand for a particular fashion brand
Population & demographics	Larger/changing population shifts demand	An ageing population increases demand for healthcare and mobility aids
Expectations of future prices	Expected price rise → demand ↑ now	Consumers rushed to buy properties before a stamp duty holiday ended
Interest rates	Lower rates → demand ↑ for credit-bought goods	Lower mortgage rates increase demand for houses

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Advertising	Effective campaigns → demand ↑	Apple's marketing increases demand for iPhones

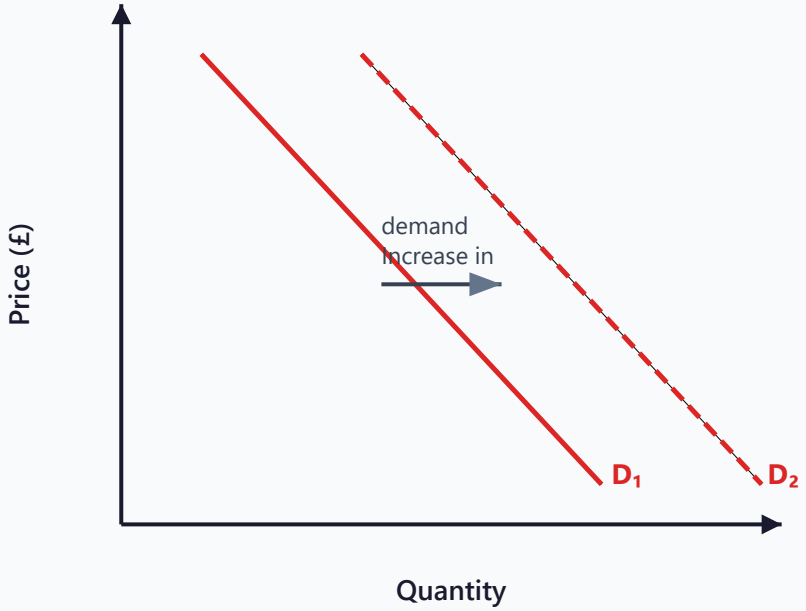

Figure 2: An increase in demand — the curve shifts to the right from D_1 to D_2 (caused by a non-price factor)

Real-World Example: The Oat Milk Boom

Between 2018 and 2023, demand for oat milk shifted dramatically to the right. This was driven by changing tastes (growing interest in plant-based diets), health concerns (lactose intolerance awareness), and environmental concerns (dairy farming's carbon footprint). The price of oat milk didn't fall — in fact it was more expensive than cow's milk — but demand surged anyway because the non-price determinants shifted the entire curve. Oatly's UK sales grew from almost nothing to over £100 million.

Supply

Supply is the quantity of a good or service that producers are *willing and able* to sell at a given

price in a given time period.

The Law of Supply

There is a **positive (direct) relationship** between price and quantity supplied, ceteris paribus. As price rises, quantity supplied rises; as price falls, quantity supplied falls. This gives the supply curve its characteristic upward slope from left to right.

Why does the supply curve slope upward?

- **Profit motive:** Higher prices mean higher profit margins, incentivising firms to produce and sell more
- **New entrants:** Higher prices attract new firms into the market, increasing total quantity supplied
- **Increasing marginal costs:** As firms expand output, they face rising costs (overtime wages, less efficient machinery, scarcer inputs). They need higher prices to justify this extra production

Figure 3: The supply curve — upward sloping, reflecting the positive relationship between price and quantity supplied

Movements Along vs Shifts of the Supply Curve

The same distinction applies to supply as to demand:

- **Movement along:** caused by a change in the good's own price (extension or contraction)
- **Shift of the curve:** caused by a change in a non-price determinant of supply

Non-Price Determinants of Supply (Supply Shifters)

Factor	Effect on Supply	Example
Costs of production	Higher costs → supply shifts left Lower costs → supply shifts right	Rising energy prices increase manufacturing costs, reducing supply of steel
Technology	Better technology → supply shifts right	Automation in car factories allows more cars to be produced at lower cost
Indirect taxes	Tax imposed → supply shifts left (higher costs for producers)	20% VAT on goods effectively raises the cost of supply
Subsidies	Subsidy → supply shifts right (lower effective costs)	Government subsidies on electric vehicles increase their supply
Number of firms	More firms → market supply shifts right	New coffee shops opening in a town increase the total supply of coffee
Weather / natural factors	Favourable → supply shifts right Adverse → supply shifts left	A drought in Brazil reduces the global supply of coffee beans
Regulation	Stricter regulation → supply shifts left	Tighter emissions standards increase car production costs

Real-World Example: Global Chip Shortage (2020–2023)

The global semiconductor shortage caused the supply curve for many goods — cars, laptops, games consoles — to shift left. COVID-19 disrupted production at chip factories in Taiwan and South Korea. Simultaneously, demand for electronics surged (people working from home). The result: prices of second-hand cars rose by 30%+, new car waiting lists stretched to 12 months, and

the PS5 was nearly impossible to buy at retail price. This perfectly illustrates how a supply shift left combined with a demand shift right creates a large price increase.

Market Equilibrium

Market equilibrium occurs where the demand curve intersects the supply curve. At this point, the quantity that consumers want to buy exactly equals the quantity that producers want to sell. There is no tendency for the price to change. The equilibrium price is also called the **market-clearing price**.

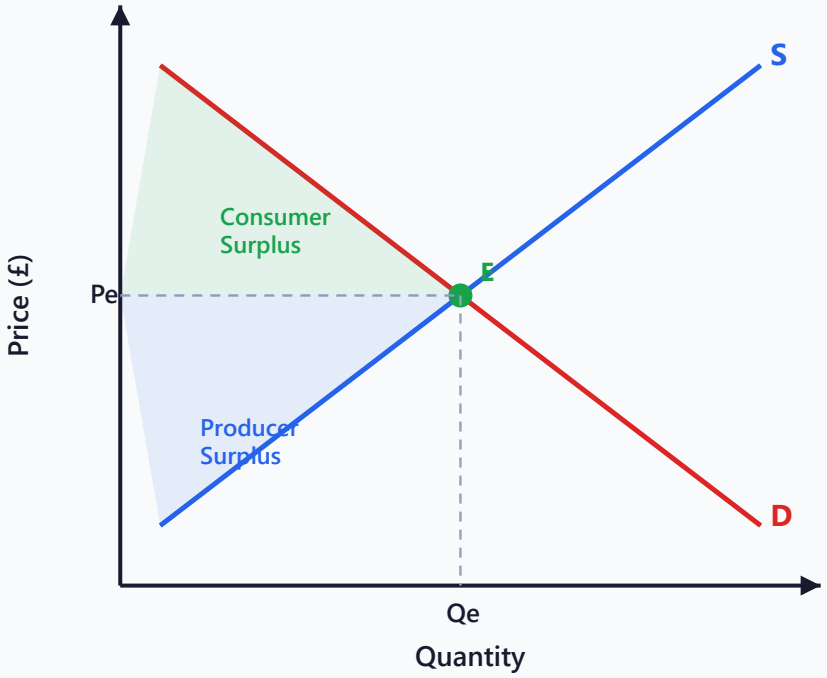

Figure 4: Market equilibrium at price Pe and quantity Qe, with consumer and producer surplus shown

How the Market Reaches Equilibrium

Markets are self-correcting through the price mechanism:

Excess Supply (Surplus) — Price Above Equilibrium

When price is above Pe, quantity supplied exceeds quantity demanded. Producers have unsold stock. To clear this surplus, firms reduce prices and cut production. As price falls, quantity demanded increases. This continues until equilibrium is restored.

Excess Demand (Shortage) — Price Below Equilibrium

When price is below Pe, quantity demanded exceeds quantity supplied. Consumers compete for limited goods — some are willing to pay more. Firms raise prices and increase production. As price rises, quantity demanded falls. This continues until equilibrium is restored.

Figure 5: At P_1 (above equilibrium) there is excess supply. At P_2 (below equilibrium) there is excess demand. Both push the market back towards equilibrium.

Changes in Equilibrium

When demand or supply curves shift, both the equilibrium price and equilibrium quantity change:

Change	Price	Quantity	Diagram

Demand increases (shifts right)	Rises ↑	Rises ↑	D shifts right along S
Demand decreases (shifts left)	Falls ↓	Falls ↓	D shifts left along S
Supply increases (shifts right)	Falls ↓	Rises ↑	S shifts right along D
Supply decreases (shifts left)	Rises ↑	Falls ↓	S shifts left along D

Real-World Example: UK Housing Market

UK house prices have risen consistently for decades. This is primarily because **demand has shifted right** (population growth, immigration, low interest rates, buy-to-let investors, foreign buyers) while **supply has barely changed** (planning restrictions, NIMBYism, limited land in desirable areas, slow construction rates). The supply of housing is highly price-inelastic in the short run — you can't build new houses overnight. So when demand keeps rising against near-fixed supply, the result is continuously rising prices. The average UK house price went from £60,000 in 1995 to over £280,000 by 2024.

Consumer Surplus and Producer Surplus

Consumer surplus (CS) is the difference between the maximum price a consumer is willing to pay and the price they actually pay. It represents the extra value or satisfaction consumers receive from a transaction. On a diagram, it is the triangular area below the demand curve and above the market price.

Producer surplus (PS) is the difference between the price a producer receives and the minimum price they would be willing to accept (their marginal cost). It represents the extra benefit producers receive. On a diagram, it is the triangular area above the supply curve and below the market price.

Total economic welfare (also called community surplus or social surplus) = consumer surplus + producer surplus. At the free-market equilibrium, total welfare is maximised — this is the condition of **allocative efficiency**.

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Anything that moves the market away from the free-market equilibrium (taxes, price controls, monopoly power) typically reduces total welfare by creating a **deadweight loss** — the welfare that is lost and not transferred to anyone.

Exam Tip: Always show consumer and producer surplus on diagrams where relevant. Examiners reward students who can identify how surpluses change when policies (taxes, subsidies, price floors/ceilings) are introduced. Shade the areas clearly and label them.

Price Elasticity of Demand (PED)

Price Elasticity of Demand (PED) measures the *responsiveness* of quantity demanded to a change in the good's own price. It answers the question: "If price changes by 1%, by what percentage does quantity demanded change?"

$$\text{PED} = \%\text{ change in quantity demanded} \div \%\text{ change in price}$$

PED is always **negative** (because of the inverse relationship between price and quantity demanded), but economists often refer to the absolute value |PED|.

Categories of PED

Value of |PED|	Term	Meaning	Example
|PED| > 1	**Price elastic**	Qd changes by a larger % than the price change. Consumers are highly responsive.	Luxury holidays, branded clothing, restaurant meals
|PED| = 1	**Unit elastic**	Qd changes by exactly the same % as the price change.	Theoretical benchmark (rare in practice)

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|PED| < 1	**Price inelastic**	Qd changes by a smaller % than the price change. Consumers are relatively unresponsive.	Petrol, bread, electricity, insulin
PED = 0	**Perfectly inelastic**	Qd does not change at all when price changes. Vertical demand curve.	Life-saving medication with no substitute
PED = −∞	**Perfectly elastic**	Any price increase causes demand to fall to zero. Horizontal demand curve.	A firm in perfect competition (many identical sellers)

Visual Comparison of Elastic vs Inelastic Demand

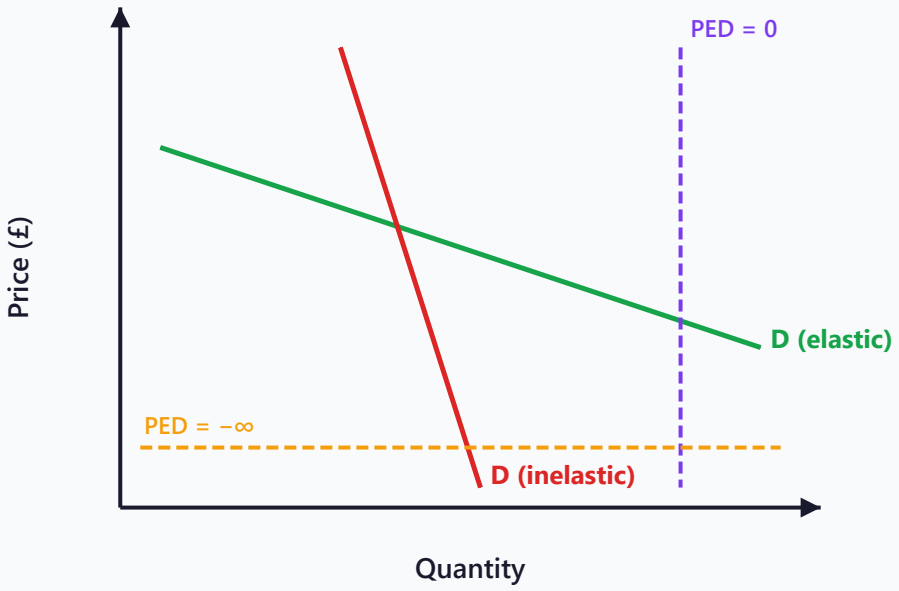

Figure 6: Comparing demand curves — flatter = more elastic, steeper = more inelastic

Determinants of PED

What makes demand more elastic or more inelastic?

Factor	More Elastic (|PED| > 1)	More Inelastic (|PED| < 1)

Substitutes	Many close substitutes available	Few or no substitutes
Necessity vs luxury	Luxury good (can live without it)	Necessity (must have it)
% of income	Large proportion of income	Small proportion of income
Time period	Long run (time to find alternatives)	Short run (locked in)
Habit / addiction	No habit formed	Strong habit or addiction
Brand loyalty	Weak brand loyalty	Strong brand loyalty
Definition of market	Narrow definition (Coca-Cola)	Broad definition (soft drinks)

Real-World Example: Petrol Prices

Petrol has highly inelastic demand in the short run. When prices spike (as they did in 2022 following Russia's invasion of Ukraine, reaching over £1.90/litre), consumers complained but kept buying because most had no immediate alternative — they still needed to drive to work. Short-run PED for petrol is estimated at around −0.2 to −0.3. However, in the long run, PED becomes more elastic (estimated −0.6 to −0.8) as people buy more fuel-efficient cars, switch to electric vehicles, use public transport, or move closer to work.

PED and Total Revenue

This is one of the most important practical applications of PED — it tells firms whether to raise or lower prices:

Elasticity	Price Increase	Price Decrease	Revenue Strategy
Elastic (|PED| > 1)	Revenue falls ↓ (lose more in quantity than gain in price)	Revenue rises ↑	Cut prices to increase revenue
Inelastic (|PED| < 1)	Revenue rises ↑ (price gain > quantity loss)	Revenue falls ↓	Raise prices to increase revenue

Unit elastic (|PED| = 1)	Revenue unchanged	Revenue unchanged	Revenue maximised at this point

Exam Tip: A simple rule: for **elastic** demand, price and revenue move in **opposite** directions. For **inelastic** demand, they move in the **same** direction. This is tested in nearly every paper.

Income Elasticity of Demand (YED)

Income Elasticity of Demand (YED) measures the responsiveness of quantity demanded to a change in consumer income. The sign (+/-) tells you the type of good; the size tells you the strength of the response.

$$\text{YED} = \%\text{ change in quantity demanded} \div \%\text{ change in income}$$

YED Value	Type of Good	What Happens When Income Rises	Examples
YED > 1 (positive, >1)	Normal good — **luxury**	Demand rises more than proportionally	Designer clothing, sports cars, fine dining, overseas holidays
0 < YED < 1 (positive, <1)	Normal good — **necessity**	Demand rises, but less than proportionally	Bread, milk, toothpaste, electricity
YED < 0 (negative)	**Inferior good**	Demand falls as income rises	Supermarket value brands, bus travel, instant noodles

Why YED Matters

For Firms

- Firms producing **luxury goods** (high positive YED) benefit greatly from economic growth — demand for their products surges during booms. But they suffer badly in recessions.

- Firms producing **inferior goods** (negative YED) are "recession-proof" — demand actually increases when the economy contracts. Discount retailers like Aldi and Lidl saw sales boom during the 2008 recession and the 2022 cost-of-living crisis.
- Understanding YED helps firms plan inventory, marketing, and investment based on economic forecasts.

For Governments

- As the economy grows, tax revenue from luxury goods (high YED) rises faster than GDP — useful for fiscal planning.
- In developing countries, as incomes rise, demand shifts from basic food towards manufactured goods and services — this drives structural economic change.

Real-World Example: Aldi & Lidl in the UK

During the 2008 financial crisis and again during the 2022 cost-of-living crisis, discount supermarkets Aldi and Lidl saw their UK market share surge. As household incomes fell in real terms, consumers switched from Waitrose and Sainsbury's (normal goods with positive YED) to budget alternatives. This is YED in action — inferior goods see rising demand when incomes fall. By 2023, Aldi had become the UK's fourth-largest supermarket.

Cross Elasticity of Demand (XED)

Cross Elasticity of Demand (XED) measures the responsiveness of quantity demanded of good A to a change in the price of good B. The sign tells you the relationship between the goods.

$$\text{XED} = \%\text{ change in quantity demanded of Good A} \div \%\text{ change in price of Good B}$$

XED Value	Relationship	Explanation	Examples
XED > 0 (positive)	**Substitutes**	A rise in the price of B causes demand for A to rise (consumers switch)	Coca-Cola & Pepsi, Uber & black cabs, butter & margarine

XED < 0 (negative)	**Complements**	A rise in the price of B causes demand for A to fall (used together)	Printers & ink, cars & petrol, phones & phone cases
XED = 0	**Unrelated goods**	Price of B has no effect on demand for A	Bread & bicycles

The **magnitude** (size) of XED indicates how closely related the goods are. A XED of +0.8 means the goods are close substitutes; a XED of +0.1 means they are weak substitutes.

Significance of XED

- **Business strategy:** If your product has a high positive XED with a competitor's product, you are vulnerable to their pricing decisions. This affects competitive strategy.
- **Complementary pricing:** Firms can sell one complementary good cheaply (e.g., printers) and make profit on the other (e.g., ink cartridges) — the "razor and blades" model.
- **Competition policy:** Regulators (like the CMA in the UK) use XED to define markets. If two goods have a high positive XED, they are in the same market — relevant for assessing mergers.

Real-World Example: The Razor-Blade Model

Gillette famously sells razors cheaply (sometimes at a loss) but charges high prices for replacement blade cartridges. The products are strong complements (high negative XED). Once a consumer buys a Gillette razor, demand for Gillette blades is price-inelastic — they're locked in. This pricing strategy exploits the complementary relationship. The same model is used by printer companies (cheap printers, expensive ink) and gaming consoles (cheap hardware, expensive games).

Price Elasticity of Supply (PES)

Price Elasticity of Supply (PES) measures the responsiveness of quantity supplied to a change in price.

$$\text{PES} = \%\text{ change in quantity supplied} \div \%\text{ change in price}$$

PES is always **positive** (because of the positive relationship between price and quantity supplied).

Determinants of PES

Factor	More Elastic Supply	More Inelastic Supply
Spare capacity	Firms have unused capacity — can increase output quickly	Firms operating at full capacity — hard to produce more
Stocks / inventories	Large stockpiles available	Cannot be stored (e.g., fresh food, services)
Time period	Long run — firms can build factories, enter markets	Short run — capacity is fixed
Factor mobility	Factors can easily switch between uses	Factors are specialised and immobile
Nature of the good	Manufactured goods (can scale production)	Agricultural goods (depend on growing seasons), natural resources
Barriers to entry	Low barriers — new firms enter quickly	High barriers — few new entrants

Exam Tip: Time is the single most important determinant of PES. In the momentary period, supply is perfectly inelastic (you can only sell existing stock). In the short run, firms can adjust variable factors (workers, raw materials). In the long run, all factors are variable — new factories can be built and new firms can enter.

Real-World Example: UK Housing Supply

The supply of housing in the UK is extremely price-inelastic, especially in London and the South East. Even when house prices doubled between 2000 and 2007, new housing construction barely increased. This is because: land is scarce (especially with planning restrictions and green belt protections), building takes years (long production period), there's a shortage of skilled

construction workers, and planning permission is slow and uncertain. This inelastic supply is a key reason why UK house prices are so volatile and so high.

Applications of Elasticity

Government Taxation

Elasticity determines the effectiveness of indirect taxes:

- Taxes on goods with **inelastic demand** (cigarettes, alcohol, petrol) raise more revenue because quantity demanded falls only slightly. This is why governments target these goods — "sin taxes" are reliable revenue generators.
- The **tax incidence** (who bears the burden) depends on relative elasticities. The more inelastic side bears more of the tax burden. If demand is very inelastic relative to supply, consumers pay most of the tax through higher prices.

Primary Commodity Markets

Agricultural goods often have both **inelastic demand** (food is a necessity) and **inelastic supply** (fixed growing seasons). This combination creates extreme price volatility:

- A small supply shock (bad harvest) causes a **large price swing**
- This creates income instability for farmers — a bumper crop can actually reduce their revenue (because demand is inelastic, the price fall more than offsets the quantity increase)

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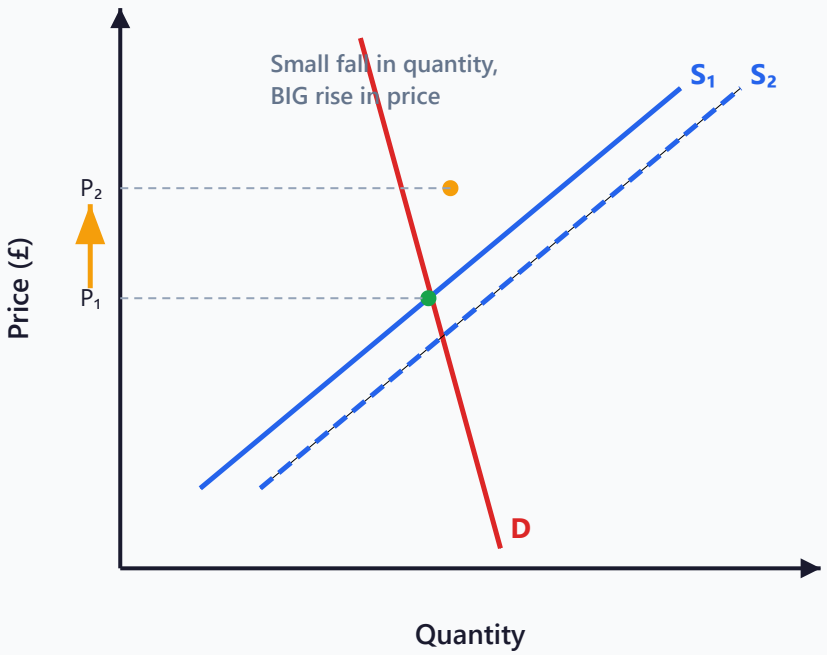

Figure 7: With inelastic demand, a supply shift left (e.g., bad harvest) causes a large price increase but only a small fall in quantity

Real-World Example: Coffee Prices in 2021

In 2021, a severe frost in Brazil (the world's largest coffee producer) destroyed a significant portion of the coffee crop. Supply shifted left sharply. Because demand for coffee is inelastic (it's a habitual daily purchase for millions), the price of arabica coffee beans nearly doubled from $1.20 to $2.50 per pound. Consumers barely reduced their consumption — they just paid more. This is exactly what the inelastic demand + supply shock model predicts.

Evaluation: Limitations of Elasticity Analysis

Critical Evaluation Points:

- **Elasticity values are estimates:** They are calculated from past data and may not accurately predict future responses. Consumer behaviour can change unpredictably.

- **PED varies along a straight-line demand curve:** A linear demand curve is elastic at the top (high price, low quantity) and inelastic at the bottom. A single PED value for the whole curve is a simplification.
- **Ceteris paribus rarely holds:** In reality, multiple factors change simultaneously, making it hard to isolate the effect of one price change.
- **Short run vs long run:** Elasticities differ significantly over time. A policy that seems effective based on short-run elasticities may have very different long-run consequences.
- **Firms may not know their elasticity:** Small businesses in particular rarely have the data or expertise to calculate elasticity precisely. They often rely on trial and error.
- **Market definition matters:** PED for "Coca-Cola" (narrow) is very different from PED for "soft drinks" (broad). How you define the market changes the elasticity.

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